

Sharp Chebyshev Atom Moduli for Fixed-Rank Subset Actions

An all-rank asymptotic theorem and exact finite orbital certificates

mxym/math research project (AI-assisted)

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Abstract

Let S_n act on the k -subsets of $[n]$, and constrain a probability law on S_n to have uniform subset-image marginals. We prove for every fixed k that the best coefficient comparing any single atom's deviation from uniform with total variation equals $1 - 2k^2/n + O_k(n^{-2})$. A corrected Chebyshev dual gives the uniform upper bound, and exactly moment-matched positive conjugation-invariant measures give the lower bound. We also prove an exact all-rank transfer-matrix cycle-compression theorem and provide independent exact finite certificates for all four-subset actions through degree 64.

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Problem and main results

For the natural action of the symmetric group S_n on $\Omega_{n,k} = \binom{[n]}{k}$, write u_n for uniform probability on S_n . A probability measure ν is admissible if

$$\nu\{\pi : \pi(A) = B\} = \binom{n}{k}^{-1} \quad \text{for all } A, B \in \Omega_{n,k}.$$

The sharp atom-vs-total-variation coefficient is

$$C_n^{(k)} = \sup_{\nu \neq u_n \text{ admissible}} \frac{|\nu(\text{id}) - 1/n!|}{\|\nu - u_n\|_{\text{TV}}}.$$

Left translation gives the same coefficient at every other permutation.

Main theorem. For every fixed $k \geq 1$,

$$C_n^{(k)} = 1 - \frac{2k^2}{n} + O_k(n^{-2}).$$

The proof below also gives the exact leading masses of the primal extremizing class mixtures and an orbital dual with a matching oscillation bound. This statement is a complete fixed-rank asymptotic theorem, not a formula for every finite (n, k) .

The finite companion determines the exact $k = 4$ coefficient for every $4 \leq n \leq 64$ with fixed rational primal-dual certificates; the separate all-rank transfer formula allows finite exact searches using only short-cycle counts.

Verification status. The full all-rank proof is analytic. Symbolic and integer checkers replay selected identities and fixed finite certificate ranges, not the universal quantified theorem. External human peer review and systematic novelty certification remain pending. The companion public proof dossier is at <https://github.com/mxym/math/tree/main/notes/sharp-robust-permanent>.

18. Universal orbital primal-dual theorem for finite permutation actions

Theorems 15–17 are instances of a general exact principle: **the sharp single-atom response to marginal-preserving perturbations is determined by a finite rational linear program on the conjugacy classes and the orbitals of the permutation representation**. In a doubly transitive action there are only two orbitals; in the two-subset action there are three. This explains the difference between the minimal-degree formula and the exact edge-action law.

Let G be a finite group acting (not necessarily transitively or faithfully) on a finite nonempty set Ω , with $e \in G$ its identity. Write u_G for the uniform law on G , and let $\mathcal{V}$ be the real vector space of signed functions $v : G \rightarrow \mathbb{R}$ satisfying

$$\sum_{g \in G} v(g) = 0, \quad \sum_{\substack{g \in G \\ g(x)=y}} v(g) = 0 \quad \text{for all } x, y \in \Omega.$$

Let $\mathcal{O}_1, \dots, \mathcal{O}_r$ be the orbitals of the action, i.e. the orbits of G on $\Omega \times \Omega$ under simultaneous relabeling. Define their *orbital displacement counts*

$$F_j(g) = \#\{x \in \Omega : (x, g(x)) \in \mathcal{O}_j\}.$$

Each F_j is constant on conjugacy classes of G . Let $C_1, \dots, C_s$ be those conjugacy classes, with $C_1 = \{e\}$, and put

$$M_{ji} = \frac{1}{|C_i|} \sum_{g \in C_i} F_j(g), \quad M \in \mathbb{Q}^{r \times s}.$$

For $\mathcal{V} \neq \{0\}$, define

$$\mathcal{C}(G, \Omega) = \sup_{\substack{v \in \mathcal{V} \\ v \neq 0}} \frac{|v(e)|}{\frac{1}{2} \sum_g |v(g)|}.$$

For $\mathcal{V} = \{0\}$, set $\mathcal{C}(G, \Omega) = 0$. The supremum is finite and $0 \leq \mathcal{C}(G, \Omega) \leq 1$.

Theorem 18 (exact orbital primal-dual characterization). The following four quantities are all equal:

$$\begin{aligned}
\mathcal{C}(G, \Omega) &= \max_{\substack{P, Q \text{ probability laws on } G \\ P\{g:gx=y\}=Q\{g:gx=y\} \ \forall x,y}} (P(e) - Q(e)) \\
&= \max_{\substack{p, q \in \mathbb{R}^s_{\geq 0} \\ \mathbf{1}^T p = \mathbf{1}^T q = 1, \ M p = M q}} (p_1 - q_1) \\
&= \min_{\lambda \in \mathbb{R}^r} \left[\max_{1 \leq i \leq s} (\mathbf{1}_{\{i=1\}} - \lambda^T M_i) - \min_{1 \leq i \leq s} (\mathbf{1}_{\{i=1\}} - \lambda^T M_i) \right] \\
&= \min_{\varphi \in \text{span}_{\mathbb{R}}\{F_1, \dots, F_r\}} \text{osc}_{g \in G} (\mathbf{1}_{\{g=e\}} - \varphi(g)).
\end{aligned}$$

Both finite LPs in the middle have **rational optimal solutions**, so $\mathcal{C}(G, \Omega) \in \mathbb{Q}$. If $\mathcal{C}(G, \Omega) > 0$, there exist two **conjugation-invariant, disjointly supported** probability laws P, Q with identical one-point marginals and $P(e) - Q(e) = \mathcal{C}(G, \Omega)$. Hence for all sufficiently small $\delta \geq 0$,

$$\nu_\delta = u_G + \delta(P - Q)$$

is a probability law with exactly the same one-point marginals as u_G and

$$\|\nu_\delta - u_G\|_{\text{TV}} = \delta, \quad \nu_\delta(e) - u_G(e) = \mathcal{C}(G, \Omega)\delta.$$

For an arbitrary probability law ν whose marginals match u_G , and any $\sigma \in G$,

$$\left| \nu(\sigma) - \frac{1}{|G|} \right| \leq \mathcal{C}(G, \Omega) \|\nu - u_G\|_{\text{TV}},$$

with this constant optimal whenever it is positive. Left translation moves the sharp construction from e to any prescribed σ .

Proof. First formulate a finite **unreduced** LP. Choose P, Q as nonnegative probability vectors indexed by G , impose their equality of all one-point marginals, and maximize $P(e) - Q(e)$. Call its optimum γ .

For any feasible pair, $v = P - Q \in \mathcal{V}$, $\|v\|_{\text{TV}} \leq 1$, so $P(e) - Q(e) \leq \mathcal{C}(G, \Omega)$. Conversely, for any nonzero $v \in \mathcal{V}$, its Jordan parts v_+, v_- each have mass $d = \|v\|_{\text{TV}} > 0$, and the probability measures $P = v_+/d$, $Q = v_-/d$ have identical marginals. Exchanging them if necessary, they give objective $|v(e)|/d$. Hence $\gamma = \mathcal{C}(G, \Omega)$. If $\mathcal{V} = \{0\}$, the two quantities are both zero.

The unreduced primal is a rational finite LP. Its dual has one unrestricted multiplier λ_{xy} for each marginal equality and two normalization multipliers a, b . Write

$$\phi(g) = \sum_{x, y \in \Omega} \lambda_{xy} \mathbf{1}_{\{g(x)=y\}}.$$

Dual feasibility is exactly

$$a \geq \mathbf{1}_{\{g=e\}} - \phi(g), \quad b \geq -\mathbf{1}_{\{g=e\}} + \phi(g) \quad \text{for all } g \in G.$$

Minimizing $a + b$ for fixed ϕ gives $\max_g(\mathbf{1}_{\{g=e\}} - \phi(g)) - \min_g(\mathbf{1}_{\{g=e\}} - \phi(g))$, its oscillation. The primal is feasible (take $P = Q = u_G$) and bounded, so **finite-dimensional LP strong duality** gives equality with the minimum oscillation. The LP has rational coefficients and a finite attained optimum; standard rational Gaussian elimination at a basic feasible optimum gives rational primal and dual certificates.

Next average an optimal primal pair under simultaneous conjugation by $h \in G$: replace P, Q by the averages of their pushforwards under $g \mapsto hgh^{-1}$. Equal one-point marginals remain equal because conjugating both pairs of image coordinates by h permutes the constraints. The identity atom and the objective are unchanged. Thus a conjugation-invariant optimal pair exists. Such pairs assign a constant probability density to each conjugacy class, and their equality of one-point marginals is equivalent to the reduced moment equation $Mp = MQ$: for a central law, each entry $\Pr(gx = y)$ is constant on the orbital containing (x, y) , and its orbital average is $\mathbb{E}F_j/|\mathcal{O}_j|$. This proves the second line of (91).

Similarly, average any dual function ϕ over conjugations. The indicator $\mathbf{1}_{\{g=e\}}$ is conjugacy invariant, while oscillation is convex and invariant under conjugation; the average cannot increase oscillation. A conjugation-averaged linear combination of the image indicators has coefficients constant on their simultaneous G -orbits in $\Omega \times \Omega$. Therefore it lies in $\text{span}\{F_j\}$. This proves the fourth line of (91). Evaluating the same oscillation on conjugacy classes gives the third line, whose entries are precisely (89).

When the optimum $\gamma > 0$, an optimal reduced pair P, Q must be **mutually singular**. Otherwise $d = \|P - Q\|_{\text{TV}} < 1$, and its Jordan parts normalized by d would form an admissible pair with strictly larger objective $\gamma/d > \gamma$, contradiction. Thus the supports of an optimal central P, Q are disjoint and (92) is a probability measure for every $0 \leq \delta \leq (|G| \max_g Q(g))^{-1}$, with all the stated identities.

Finally, for any law ν with the uniform law's one-point marginals, $v = \nu - u_G \in \mathcal{V}$ and the defining norm bound gives (94) at the identity. Replacing ν by its left translate by σ^{-1} preserves marginal equality with u_G , total variation and atom excess, proving (94) for every σ . The same translation transports the attained equality. QED.

Consequences and scope. Theorem 18 gives a fully finite **exact certificate interface** for *every* finite permutation group action: exhibit a central primal pair of matching orbital moments and a dual orbital linear combination whose oscillation equals the primal atom excess. It also explains Theorem 15 (rank-two orbital geometry and a single fixed-point statistic), Theorem 16 (rank-three geometry in degree ten), and Theorem 17 (a full parity-dependent family of exact rank-three optima). The theorem does **not** assert a similarly explicit symbolic formula for arbitrary higher-rank actions. Determining a closed analytic optimum for S_n acting on k -subsets with $k \geq 3$ is a natural next target; numerical LP output there must be converted to rational primal-dual certificates before any theorem claim.

23. General k -subset orbital Bernstein limits and a Chebyshev research direction

The first-order constants 2, 8, 18 in the one-, two-, and three-subset actions have a common structure. The orbital basis of **every fixed subset rank** converges to the Bernstein polynomial

basis, while the explicit duals in ranks 1, 2, 3 converge to *shifted Chebyshev polynomials*. The first observation is a rigorously proved general theorem. Its extension to a sharp all- k atom-TV asymptotic remains a conjecture.

Let $1 \leq k \leq n$, let $\Omega_{n,k} = \binom{[n]}{k}$, and define

$$F_j^{(k)}(g) = \#\{E \in \Omega_{n,k} : |E \cap g(E)| = j\} \quad (0 \leq j \leq k).$$

Write $x(g)$ for the number of fixed vertices and $a(g) = x(g)/n$. Define the Bernstein basis $B_{j,k}(a) = \binom{k}{j} a^j (1-a)^{k-j}$.

Theorem 23 (uniform Bernstein orbital approximation). For every $n \geq \max(k, 2)$, every permutation $g \in S_n$, and every $0 \leq j \leq k$,

$$\left| \frac{F_j^{(k)}(g)}{\binom{n}{k}} - B_{j,k}(a(g)) \right| \leq \frac{k(k-1)}{n-1} + \frac{k(k-1)}{2n}.$$

Indeed the **total variation distance between the entire two distributions** on $j = 0, \dots, k$ satisfies the same bound. Consequently for every fixed k , the $k+1$ normalized orbital statistics converge uniformly over $g \in S_n$, at rate $O_k(n^{-1})$, to the Bernstein basis of polynomials of degree at most k .

Proof. Choose a uniformly random k -subset E , and let $X = |E \cap g(E)|$. Let S be the fixed-vertex set of g , with $|S| = x$, and put $Y = |E \cap S|$. Every fixed vertex in E also belongs to $g(E)$, hence $X \geq Y$. Any extra element $v \in E \cap g(E) \setminus S$ requires the two **distinct** vertices v and $g^{-1}(v)$ both to belong to E . For each moved vertex v , the probability of this pair event is exactly $k(k-1)/(n(n-1))$. By the union bound,

$$\mathbb{P}(X \neq Y) \leq (n-x) \frac{k(k-1)}{n(n-1)} \leq \frac{k(k-1)}{n-1}.$$

The variable Y has the hypergeometric distribution of the number of successes in k draws without replacement from a population of n with x successes. Draw instead k vertices independently and uniformly with replacement, and let $Z \sim \text{Binomial}(k, x/n)$ count successes. The distribution of the ordered independent sample **conditioned on distinctness** is exactly that of ordered sampling without replacement. The probability of a collision is at most $\binom{k}{2}/n = k(k-1)/(2n)$; thus the total variation distance between Y and Z is at most this probability. By the coupling characterization and the triangle inequality,

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(Z)) \leq \mathbb{P}(X \neq Y) + d_{\text{TV}}(\mathcal{L}(Y), \mathcal{L}(Z)),$$

which is (123) for the full distributions and hence for each coordinate. QED.

Proposition 24 (Chebyshev limiting duals in ranks 1, 2, 3). Let T_k denote the Chebyshev polynomial of the first kind, $T_k(\cos \theta) = \cos(k\theta)$. In each of the already proved ranks $k = 1, 2, 3$, the leading nonidentity orbital dual polynomial of the sharp or sharp-order certificates is

$$H_k(a) = \frac{1 - T_k(2a - 1)}{2},$$

namely

$$H_1(a) = 1 - a, \quad H_2(a) = 4a(1 - a), \quad H_3(a) = (1 - a)(4a - 1)^2.$$

All satisfy $0 \leq H_k(a) \leq 1$ for $0 \leq a \leq 1$, $H_k(1) = 0$, and the endpoint derivative identity

$$-H'_k(1) = k^2.$$

Proof. The polynomial identities follow by substituting $T_1(t) = t$, $T_2(t) = 2t^2 - 1$, and $T_3(t) = 4t^3 - 3t$. For $k = 1$, the fixed-point dual $h(g) = \mathbf{1}_{\{g=e\}} + (n - x(g))/n$ has nonidentity profile $H_1(x/n)$ exactly. For $k = 2$, the exact duals (76) and (81) have the common leading nonidentity profile $4a(1 - a)$: substitute the leading terms of (68) with $x = an$ and $y = O(n)$. For $k = 3$, equation (113) provides the exact leading polynomial H_3 . The range and derivative statements follow from $|T_k(t)| \leq 1$ on $[-1, 1]$ and $T'_k(1) = k^2$. QED.

This exhibits why the exact first-order constants are

k	$\lim_{n \rightarrow \infty} n(1 - C_n^{(k)})$
1	2 (Theorem 15),
2	8 (Theorem 17),
3	18 (Theorem 22).

A nonidentity vertex permutation must move at least two vertices, so the closest possible fixed-point fraction to 1 is $1 - 2/n$. For a shifted Chebyshev dual, the endpoint loss is therefore $2k^2/n + O_k(n^{-2})$. This interpretation is exact for the three established ranks; it motivates but **does not prove** the following general question.

Historical conjecture (now proved in Theorem 25). For every fixed $k \geq 4$, the optimal coefficient for S_n acting on its k -subsets satisfies

$$C_n^{(k)} = 1 - \frac{2k^2}{n} + O_k(n^{-2}) \quad (n \rightarrow \infty).$$

A viable proof must address **both sides**: construct a uniformly valid orbital dual with the appropriate subleading corrections, and match it by genuine nonnegative class measures with exactly equal k -subset image marginals. Theorem 23 alone proves only Bernstein convergence; it does not control the $1/n$ coefficient or justify a Chebyshev optimizer. Numerical LP output at fixed degrees also cannot establish (128). **Status update:** this was the research conjecture at the time of Section 23; the full fixed-rank statement is now **proved** in Theorem 25, Section 24. The finite- n exact classification remains open.

24. Resolution of the fixed-rank Chebyshev atom-modulus conjecture

The conjecture in Section 23 admits a **complete affirmative proof for every fixed subset rank**. The argument has three independent parts: a uniform first-order orbital expansion, a corrected Chebyshev dual giving an upper bound, and genuine positive probability measures at Chebyshev–Lobatto nodes giving the matching lower bound. None of these uses the finite LP computations as a proof premise.

For fixed $k \geq 1$, denote by $C_n^{(k)}$ the optimal single-atom versus total-variation coefficient for the natural S_n -action on $\binom{[n]}{k}$, under preservation of every one-point subset-image marginal, as defined by Theorem 18.

Theorem 25 (sharp universal fixed-rank law). For every fixed integer $k \geq 1$,

$$C_n^{(k)} = 1 - \frac{2k^2}{n} + O_k(n^{-2}) \quad (n \rightarrow \infty).$$

In particular,

$$\lim_{n \rightarrow \infty} n(1 - C_n^{(k)}) = 2k^2.$$

Both sides of the asymptotic bound follow from independently specified, rigorous constructions. The primal measures have **exactly** matching subset-image marginals, not merely asymptotically matching ones.

24.1. A uniform first-order orbital expansion

For $n \geq 2k$, $g \in S_n$, let x and y be its numbers of 1-cycles and 2-cycles, and set $a = x/n$, $b = y/n$. Necessarily $0 \leq a \leq 1$ and $0 \leq b \leq (1 - a)/2$. Choose uniformly $E \in \binom{[n]}{k}$, put $X = |E \cap g(E)|$, and write

$$G_{n,g}(t) = \mathbb{E}[t^X], \quad B(t) = 1 - a + at.$$

Lemma 26 (uniform orbital generating expansion). For each fixed $k \geq 2$,

$$\begin{aligned} G_{n,g}(t) = B(t)^k + \frac{k(k-1)}{n} B(t)^{k-2} & \left[(1-a)(t-1) \right. \\ & \left. + (b - \tfrac{1}{2}a(1-a))(t-1)^2 \right] + O_k(n^{-2}). \end{aligned}$$

The remainder is **uniform in every permutation** $g \in S_n$, and is bounded coefficientwise as a polynomial in t , with a constant depending only on fixed k . For $k = 1$, $G_{n,g}(t) = 1 - a + at$ exactly.

Proof. Let S be the fixed-vertex set, and put $Y = |E \cap S|$. This is hypergeometric, and

$$\mathbb{E}(1+u)^Y = \sum_{j=0}^k \binom{k}{j} \frac{(x)_j}{(n)_j} u^j.$$

For fixed $j \leq k$ and all integers $0 \leq x \leq n$, the falling-factorial ratio has the uniform expansion

$$\frac{(x)_j}{(n)_j} = a^j - \frac{\binom{j}{2}}{n} a^{j-1} (1-a) + O_k(n^{-2});$$

the apparent a^{-1} singularity does not occur because $j \geq 2$ in the correction. This follows by multiplying out the fixed-degree falling factorial polynomials and using $(n)_j = n^j(1 - \binom{j}{2}/n + O_k(n^{-2}))$ for $n \geq 2k$. Summing over j gives

$$\mathbb{E}[t^Y] = B(t)^k - \frac{k(k-1)}{2n} a(1-a)(t-1)^2 B(t)^{k-2} + O_k(n^{-2}).$$

Now $X - Y$ counts moved vertices $v \in E$ for which $g^{-1}(v) \in E$. In cycles of length at least three, the directed edges $g^{-1}v \rightarrow v$ give $L = n - x - 2y$ *distinct unordered selected-pair events*, each of which adds **one** to $X - Y$. A transposition gives only one unordered pair event, but when selected, contributes **two** to $X - Y$. There are y such events. Each individual pair is contained in E with probability

$$\frac{k(k-1)}{n(n-1)} = \frac{k(k-1)}{n^2} + O_k(n^{-3}).$$

Distinct event edges form a graph of maximum vertex degree two. There are $O(n)$ pairs of event edges that share a vertex; selecting their three distinct endpoints has probability $O_k(n^{-3})$. There are $O(n^2)$ disjoint event-edge pairs; selecting their four endpoints has probability $O_k(n^{-4})$. Therefore the probability of **two or more distinct pair events** is $O_k(n^{-2})$, uniformly in g .

Conditioned on a particular selected pair, both its vertices are moved, and the remaining $k-2$ chosen vertices form a uniform $(k-2)$ -subset of the other $n-2$ vertices. Their fixed-point count has generating polynomial $B(t)^{k-2} + O_k(n^{-1})$ coefficientwise, by the same finite hypergeometric approximation. Discarding the multiple-event configurations, whose contribution is coefficientwise $O_k(n^{-2})$, gives

$$\begin{aligned} \mathbb{E}[t^X] - \mathbb{E}[t^Y] &= \frac{k(k-1)}{n} B(t)^{k-2} [(1-a-2b)(t-1) + b(t^2-1)] \\ &\quad + O_k(n^{-2}) \\ &= \frac{k(k-1)}{n} B(t)^{k-2} [(1-a)(t-1) + b(t-1)^2] + O_k(n^{-2}). \end{aligned}$$

The coefficientwise estimates are legitimate because k is fixed, $X \leq k$, and any sum of selected-edge indicators on a k -set is bounded by k . Combining (132) and (133) proves (131). QED.

Every polynomial $P(a)$ of degree at most k has a unique Bernstein representation

$$P(a) = \sum_{j=0}^k \beta_j \binom{k}{j} a^j (1-a)^{k-j}.$$

Define

$$\mathcal{L}_{n,g}(P) = \sum_{j=0}^k \beta_j \frac{F_j^{(k)}(g)}{\binom{n}{k}}.$$

Corollary 27 (universal first correction). For each fixed $k \geq 2$,

$$\mathcal{L}_{n,g}(P) = P(a) + \frac{J_P(a, b)}{n} + O_{k,P}(n^{-2}),$$

uniformly in g , where

$$J_P(a, b) = (k-1)(1-a)P'(a) + \left(b - \frac{3}{2}a(1-a)\right)P''(a).$$

Proof. Apply the coefficient functional $t^j \mapsto \beta_j$ to (131). The Bernstein differentiation identities give

$$\sum_j \beta_j [t^j] ((t-1)^2 B(t)^{k-2}) = \frac{P''(a)}{k(k-1)}$$

and

$$\sum_j \beta_j [t^j] ((t-1)B(t)^{k-2}) = \frac{P'(a)}{k} - \frac{aP''(a)}{k(k-1)}.$$

The asserted expression (135) follows directly, retaining the uniform coefficientwise error. QED.

24.2. An explicitly corrected Chebyshev dual for every rank

For each integer $k \geq 2$, let

$$H(a) = \frac{1 - T_k(2a-1)}{2}, \quad a_j = \frac{1 + \cos(j\pi/k)}{2} \quad (j = 0, \dots, k),$$

where T_k is the first Chebyshev polynomial. Then

$$1 = a_0 > a_1 > \dots > a_k = 0, \quad H(a_j) = \begin{cases} 0, & j \text{ even}, \\ 1, & j \text{ odd}, \end{cases} \quad H'(1) = -k^2.$$

All interior nodes $a_1, \dots, a_{k-1}$ are nondegenerate extrema: $H''(a_j) < 0$ at odd j and $H''(a_j) > 0$ at even j . At the left endpoint,

$$H'(0) = (-1)^k k^2, \quad \operatorname{sgn} H''(0) = (-1)^{k+1} \quad (k \geq 2).$$

Put $\rho = 2k^2$. Let J_H be the correction (135) with $P = H$. Prescribe

$$b_j = 0 \quad (1 \leq j < k), \quad b_k = \frac{1}{2}, \quad \tau_j = \begin{cases} 0, & j \text{ odd}, \\ \rho, & j \text{ even}. \end{cases}$$

There is a **unique polynomial R of degree at most k** satisfying

$$R(1) = 0, \quad R(a_j) = \tau_j - J_H(a_j, b_j) \quad (1 \leq j \leq k),$$

because these specify its values at $k + 1$ distinct nodes. Define

$$J(a, b) = J_H(a, b) + R(a).$$

The first-correction contacts are therefore

$$J(a_j, 0) = \tau_j \quad (1 \leq j < k), \quad J(0, \tfrac{1}{2}) = \tau_k, \quad J(1, 0) = 0, \quad \partial_b J(a, b) = H''(a).$$

Write the degree- k Bernstein coefficients of H, R as $\beta_0, \dots, \beta_k$ and $\gamma_0, \dots, \gamma_k$. Since $H(1) = R(1) = 0$, one has $\beta_k = \gamma_k = 0$. Define the **finite, explicit orbital dual function**

$$h_{n,k}(g) = \mathbf{1}_{\{g=e\}} + \sum_{j=0}^{k-1} \left(\beta_j + \frac{\gamma_j}{n} \right) \frac{F_j^{(k)}(g)}{\binom{n}{k}}.$$

At the identity $h_{n,k}(e) = 1$ **exactly**. By Corollary 27, for every nonidentity permutation, with $a = x(g)/n$, $b = y(g)/n$,

$$h_{n,k}(g) = H(a) + \frac{J(a, b)}{n} + O_k(n^{-2})$$

uniformly over the entire feasible region $0 \leq a \leq 1 - 2/n$, $0 \leq b \leq (1 - a)/2$.

Lemma 28 (uniform corrected-dual bound). For each fixed $k \geq 2$, there is $D_k < \infty$ such that for all sufficiently large n and all nonidentity $g \in S_n$,

$$\boxed{\frac{\rho}{n} - \frac{D_k}{n^2} \leq h_{n,k}(g) \leq 1 + \frac{D_k}{n^2}}.$$

Proof. By (144), it suffices to prove the claim for $H(a) + J(a, b)/n$; the uniform remainder can be absorbed into D_k . We cover the compact feasible parameter region by neighborhoods of the finitely many extrema of H and their complement. Every constant below may depend on fixed k , but not on n, a, b .

Interior maxima. For odd $1 \leq j < k$, choose a neighborhood of a_j on which $H'' < 0$ and $H(a) \leq 1 - c(a - a_j)^2$ for some $c > 0$. By (142), $J(a, b) \leq J(a, 0)$ for $b \geq 0$. Since $J(a_j, 0) = 0$, smoothness gives $J(a, 0) \leq M|a - a_j|$. Completing the square,

$$H(a) + J(a, b)/n \leq 1 - c(a - a_j)^2 + \frac{M|a - a_j|}{n} \leq 1 + \frac{M^2}{4cn^2}.$$

The lower bound is automatic nearby because H stays bounded above zero.

Interior minima. For even $1 \leq j < k$, choose a neighborhood on which $H'' > 0$ and $H(a) \geq c(a - a_j)^2$. Then $J(a, b) \geq J(a, 0) \geq \rho - M|a - a_j|$, by (142) and the contact $J(a_j, 0) = \rho$. Thus

$$H(a) + J(a, b)/n \geq c(a - a_j)^2 + \frac{\rho - M|a - a_j|}{n} \geq \frac{\rho}{n} - \frac{M^2}{4cn^2}.$$

The upper bound is automatic because H stays away from one.

Endpoint $a = 0$. If k is odd, $H(0) = 1$, $H'(0) = -k^2$, $H''(0) > 0$. Near zero, $J(a, b) \leq J(a, (1 - a)/2)$, because $b \leq (1 - a)/2$ and the slope in b is positive. The latter function vanishes at $a = 0$ by (142), so is at most Ma ; meanwhile $H(a) \leq 1 - ca$. Hence the upper bound holds for large n , with the lower bound automatic.

If k is even, $H(0) = 0$, $H'(0) = k^2$, $H''(0) < 0$. Then $J(a, b) \geq J(a, (1 - a)/2) \geq \rho - Ma$ near zero. Since $H(a) \geq ca$, the lower bound holds for large n , while the upper bound is automatic.

Endpoint $a = 1$. Here $H(1) = 0$, $H'(1) = -k^2$, and $J(1, 0) = 0$. Put $s = 1 - a$, so $b \leq s/2$, and choose M large enough that for all sufficiently small $s \geq 0$,

$$H(1 - s) \geq k^2s - Ms^2, \quad J(1 - s, b) \geq -Ms.$$

For nonidentity permutations $s \geq 2/n$. Choose a fixed small neighborhood and then n large enough that $f_n(s) = k^2s - Ms^2 - Ms/n$ is **increasing** there. Thus

$$H(a) + J(a, b)/n \geq f_n(2/n) = \frac{2k^2}{n} + O_k(n^{-2}).$$

The upper bound is automatic near this zero of H .

Compact complement. Away from all these finitely many nodes, continuity gives a constant $\eta > 0$ with $\eta \leq H(a) \leq 1 - \eta$. The polynomial J is bounded uniformly on the compact feasible triangle. Both desired inequalities hold with fixed slack for all large n .

Together these neighborhoods cover all $g \neq e$, proving (145). QED.

By Theorem 18, the sharp coefficient is no greater than the **oscillation** of any dual function consisting of the identity indicator plus a linear combination of orbital statistics. Since $h_{n,k}(e) = 1$, (145) implies

$$\boxed{C_n^{(k)} \leq 1 - \frac{2k^2}{n} + O_k(n^{-2}).}$$

The case $k = 1$ is already given exactly by Theorem 15: $C_n^{(1)} = (n - 2)/n$.

24.3. Matching exact positive probability measures

We now obtain the reverse inequality in (129), for **every sufficiently large** n and fixed k . Crucially, this constructs genuinely nonnegative probability measures whose subset-image marginals agree **exactly**, not just in an asymptotic expansion.

Let $a_0 = 1 > a_1 > \dots > a_k = 0$ be the nodes (136). For each $1 \leq j \leq k$, choose an integer $x_j(n)$ with

$$|x_j(n) - na_j| \leq 1.$$

Let $g_{j,n} \in S_n$ have exactly $x_j(n)$ fixed vertices and one additional cycle of length $n - x_j(n)$. Because k is fixed and every $a_j < 1$, for sufficiently large n all these long cycles have length at least $k + 1$. The k cycle types are mutually distinct, nonidentity and not transpositions. Let I_n denote the identity, and T_n a transposition.

Define the **rational orbital vector** in k coordinates

$$V_n(g) = \left(\frac{F_0^{(k)}(g)}{\binom{n}{k}}, \dots, \frac{F_{k-1}^{(k)}(g)}{\binom{n}{k}} \right).$$

Thus $V_n(I_n) = 0$. Form the $k \times k$ matrix D_n with columns

$$D_{n,j} = \begin{cases} V_n(g_{j,n}) - V_n(I_n), & j \text{ odd}, \\ V_n(T_n) - V_n(g_{j,n}), & j \text{ even}, \end{cases} \quad (1 \leq j \leq k),$$

and consider the **specified rational linear system**

$$D_n w_n = V_n(T_n) - V_n(I_n), \quad w_n = (w_{1,n}, \dots, w_{k,n})^T.$$

We prove that D_n is invertible, that **all** weights $w_{j,n}$ are strictly positive, and that

$$\sum_{j \text{ odd}} w_{j,n} = \frac{2k^2}{n} + O_k(n^{-2})$$

for all sufficiently large n .

Write $B_j(a) = \binom{k}{j} a^j (1-a)^{k-j}$ and $\mathbf{B}(a) = (B_0(a), \dots, B_{k-1}(a))$. The uniform Bernstein approximation in Theorem 23, together with the rounding of $x_j(n)$, gives

$$V_n(g_{j,n}) = \mathbf{B}(a_j) + O_k(n^{-1}).$$

The exact transposition count is

$$\Pr_{E \in \binom{[n]}{k}} (|E \cap T_n(E)| = k-1) = \frac{2 \binom{n-2}{k-1}}{\binom{n}{k}} = \frac{2k(n-k)}{n(n-1)} = \frac{2k}{n} + O_k(n^{-2}),$$

and every other changed-image orbital has probability zero. Equivalently,

$$V_n(T_n) - V_n(I_n) = -\frac{2}{n}\mathbf{B}'(1) + O_k(n^{-2}).$$

The limiting matrix D_∞ has columns $+\mathbf{B}(a_j)$ at odd j and $-\mathbf{B}(a_j)$ at even j . It is invertible: every $B_i(a)$ with $0 \leq i < k$ contains the factor $(1 - a)$. Dividing the evaluation matrix at the k distinct points $a_1, \dots, a_k < 1$ by these nonzero row factors leaves an evaluation matrix for a basis of polynomials of degree at most $k - 1$, which is invertible by the Vandermonde theorem. Therefore D_n is invertible for all large n , and its inverses are uniformly bounded in n .

For each $j = 0, \dots, k$, let $\ell_j(a)$ be the degree- k **Lagrange cardinal polynomial** for the nodes $a_0, \dots, a_k$. Thus for every polynomial P of degree at most k ,

$$P'(1) = \sum_{j=0}^k \ell'_j(1)P(a_j).$$

For $j \geq 1$ the sign of $\ell'_j(1)$ is $(-1)^j$: indeed

$$\ell'_j(1) = \frac{\prod_{i \notin \{0, j\}} (1 - a_i)}{\prod_{i \neq j} (a_j - a_i)},$$

and exactly j denominator factors are negative, since the nodes are strictly decreasing. Put

$$u_j = 2|\ell'_j(1)| > 0, \quad 1 \leq j \leq k.$$

Applying (153) to $B_0, \dots, B_{k-1}$, all of which vanish at $a_0 = 1$, shows that u solves the **limiting** equation

$$D_\infty u = -2\mathbf{B}'(1).$$

Now $D_n = D_\infty + O_k(n^{-1})$ from (151) and $n[V_n(T_n) - V_n(I_n)] = -2\mathbf{B}'(1) + O_k(n^{-1})$ from (152). Uniform boundedness of D_n^{-1} therefore gives

$$\boxed{w_{j,n} = \frac{u_j}{n} + O_k(n^{-2}) \quad (1 \leq j \leq k).}$$

Since every $u_j > 0$, this proves **strict positivity of the exact rational solution** $w_{j,n}$ for sufficiently large n .

Finally, the Chebyshev polynomial H has $H(a_j) = 1$ for odd j and 0 for even j , while $H(1) = 0$. Applying (153) to H and using $H'(1) = -k^2$,

$$\sum_{j \text{ odd}} u_j = -2 \sum_{j \text{ odd}} \ell'_j(1) = -2H'(1) = 2k^2.$$

Equation (150) follows.

Now define actual **central probability laws**

$$P_n = \left(1 - \sum_{j \text{ odd}} w_{j,n}\right) U_{\{I_n\}} + \sum_{j \text{ odd}} w_{j,n} U_{[g_{j,n}]},$$

$$Q_n = \left(1 - \sum_{j \text{ even}} w_{j,n}\right) U_{[T_n]} + \sum_{j \text{ even}} w_{j,n} U_{[g_{j,n}]},$$

where $U_{[g]}$ is uniform probability on the conjugacy class of g . By (155), every coefficient is positive for sufficiently large n ; the two supports are disjoint.

Equation (149) is **exact** and says that all k orbital coordinates of P_n and Q_n match. The final orbital coordinate matches automatically because the $k + 1$ normalized coordinates sum to one. Since the measures are conjugation-invariant, equality of all orbital moments is equivalent to equality of **every** k -subset one-point image marginal (Theorem 18).

For every sufficiently small rational $\delta > 0$, the signed perturbation $\nu_\delta = u_{S_n} + \delta(P_n - Q_n)$ is therefore a nonnegative probability measure with **exact uniform subset-image marginals**, total variation exactly δ , and

$$\nu_\delta(I_n) - \frac{1}{n!} = \delta P_n(I_n) = \delta \left(1 - \frac{2k^2}{n} + O_k(n^{-2})\right).$$

Consequently Theorem 18 supplies the reverse estimate

$$\boxed{C_n^{(k)} \geq 1 - \frac{2k^2}{n} + O_k(n^{-2}).}$$

Together with the matching dual bound (146), this proves Theorem 25 for every fixed $k \geq 2$. For $k = 1$, the already proved exact identity $C_n^{(1)} = (n - 2)/n$ finishes the statement. **Theorem 25 is completely proved. QED.**

24.4. Exact four-subset example and the remaining finite-degree problem

In the first previously unresolved rank $k = 4$,

$$H(a) = 16a(1 - a)(2a - 1)^2,$$

and the interpolation system (140) yields the explicitly checkable **integer-coefficient correction**

$$R(a) = 16(a - 1)(200a^3 - 232a^2 + 63a - 4).$$

Their degree-four Bernstein coefficient vectors are

$$(\beta_0, \dots, \beta_4) = (0, 4, -16/3, 4, 0),$$

$$(\gamma_0, \dots, \gamma_4) = (64, -204, 944/3, -108, 0).$$

The published exact checker in `code/check_k4_chebyshev_dual.py` independently verifies the first-order orbital expansion, every Chebyshev contact, and the Bernstein coefficients using algebraic arithmetic in $\mathbb{Q}(\sqrt{2})$; it additionally constructs **strictly positive rational primal weights** at several larger finite degrees and checks the moment equations exactly.

Precisely what is now closed. Equation (129) settles the sharp first-order asymptotic coefficient for **every fixed** k , not merely $k = 1, 2, 3$, and proves the higher-rank conjecture previously stated in Section 23. It does **not** determine the individual exact value of $C_n^{(k)}$ for arbitrary finite n, k ; nor does the qualitative proof specify a common explicit threshold in n beyond which all primal weights are positive. Those stronger effective/finite problems remain open, and no novelty or external-peer-review claim is implied.

24.5. Explicit asymptotic masses at every Chebyshev node

The construction in (149)–(157) admits closed, positive leading weights, not merely an existence argument.

Corollary 29 (universal Chebyshev–Lobatto mass formula). In Theorem 25, define $c_j = 1$ for $1 \leq j < k$ and $c_k = 2$. For the nodes $a_j = (1 + \cos(j\pi/k))/2$, the unique asymptotic solution to (149) has

$$w_{j,n} = \frac{1}{n} \frac{4}{c_j(1 - a_j)} + O_k(n^{-2}) = \frac{1}{n} \frac{8}{c_j[1 - \cos(j\pi/k)]} + O_k(n^{-2}) \quad (1 \leq j \leq k).$$

Consequently the constructed central primal measures satisfy

$$P_n(I_n) = 1 - \frac{2k^2}{n} + O_k(n^{-2}), \quad Q_n([T_n]) = 1 - \frac{2(k^2 - 1)}{3n} + O_k(n^{-2}),$$

where $Q_n([T_n])$ denotes the total mass of the transposition conjugacy class. In particular, $w_{k,n} = 2/n + O_k(n^{-2})$ for every k , and the entire leading mass profile is **explicit**.

Proof. For Chebyshev–Lobatto nodes in decreasing order, the Lagrange barycentric weights are proportional to $(-1)^j/c_j$, with both endpoint denominators equal to 2 and all interior denominators equal to 1. The derivative formula for a Lagrange cardinal polynomial at $a_0 = 1$ is therefore

$$\ell'_j(1) = \frac{(-1)^j 2}{c_j(1 - a_j)}, \quad j \geq 1.$$

Combining with (154)–(155) gives (161), including the endpoint mass $w_{k,n} = 2/n + O_k(n^{-2})$.

The odd-weight identity is already (156). For the even weights, interpolation of the complementary polynomial $1 - H$ gives

$$\sum_{\substack{j \geq 2 \\ j \text{ even}}} \ell'_j(1) = (1 - H)'(1) - \ell'_0(1) = k^2 - \ell'_0(1).$$

But $\ell'_0(1) = \sum_{j=1}^k (1 - a_j)^{-1} = (2k^2 + 1)/3$. For completeness, the node polynomial for $a_1, \dots, a_k$, under $t = 2a - 1$, is proportional to $(t + 1)U_{k-1}(t)$, where U_{k-1} is the Chebyshev polynomial of the second kind. Logarithmic differentiation at $t = 1$, using $U_{k-1}(1) = k$ and $U'_{k-1}(1) = k(k^2 - 1)/3$, gives exactly $\ell'_0(1) = 1 + 2(k^2 - 1)/3 = (2k^2 + 1)/3$. Hence

$$\sum_{j \text{ even}} u_j = 2 \left(k^2 - \frac{2k^2 + 1}{3} \right) = \frac{2(k^2 - 1)}{3},$$

which, with (155), yields the second formula in (162). QED.

The weights in (161) are the *leading asymptotics* of strictly positive, **exactly rational** solutions of (149). They are not claimed to be the exact finite- n weights at all n , and not every leading coefficient is rational.

25. An exact cycle-index and transfer-matrix compression theorem for every rank

The earlier rank-three compression theorem (Theorem 20) is a special case of a general exact identity. For **every** fixed subset size k , the full orbital vector of a permutation depends only on its cycle counts of lengths **at most** k . This observation is stronger than the first-order Bernstein approximation: it is an identity over integers for **every finite** n and provides an exact proof/certificate interface for higher-rank finite classifications.

Write $c_\ell(g)$ for the number of length- ℓ cycles of $g \in S_n$. Let u, t be commuting formal variables and define the 2×2 transfer matrix

$$M(u, t) = \begin{pmatrix} 1 & u \\ 1 & ut \end{pmatrix}.$$

Let $\lambda_+(u, t) \in \mathbb{Q}[t][[u]]$ be the unique formal-power-series root with constant coefficient 1 of

$$\lambda^2 - (1 + ut)\lambda + u(t - 1) = 0.$$

Thus $\lambda_+ = 1 + u + (t - 1)u^2 + \dots$. Define the **complete rank- k orbital polynomial**

$$\mathcal{F}_{n,k,g}(t) = \sum_{j=0}^k F_j^{(k)}(g)t^j.$$

Theorem 30 (exact all-rank cycle compression). For every integer $1 \leq k \leq n$,

$$\mathcal{F}_{n,k,g}(t) = [u^k] \left\{ \lambda_+(u, t)^{n - \sum_{\ell=1}^k \ell c_\ell(g)} \prod_{\ell=1}^k (\text{tr } M(u, t)^\ell)^{c_\ell(g)} \right\}.$$

Only the coefficients through u^k of the formal power series are required. In particular, **if two permutations have the same $c_1, \dots, c_k$, then all their k -subset orbital statistics agree exactly**:

$$c_\ell(g) = c_\ell(h) \ (1 \leq \ell \leq k) \implies F_j^{(k)}(g) = F_j^{(k)}(h) \ (0 \leq j \leq k).$$

Furthermore, nonnegative integers $c_1, \dots, c_k$ arise from a permutation of n points if and only if

$$r = n - \sum_{\ell=1}^k \ell c_\ell \in \{0\} \cup \{k+1, k+2, \dots\}.$$

Consequently the general orbital primal-dual program of Theorem 18 can be compressed **without any loss in optimality** to at most $O_k(n^k)$ short-cycle-count vectors, instead of enumerating every integer partition of n . Every feasible vector has a canonical representative of cycle type $1^{c_1} \dots k^{c_k} r$ when $r \geq k+1$, or $1^{c_1} \dots k^{c_k}$ when $r = 0$.

Proof. Restrict g to a cycle of length ℓ , written as the cyclic vertex list $v_1, \dots, v_\ell$. Choosing a subset of its vertices is equivalent to choosing a cyclic binary word $\epsilon = (\epsilon_1, \dots, \epsilon_\ell)$, where $\epsilon_i = 1$ means v_i is selected. Its contribution to the size of the selected set is $\sum_i \epsilon_i$; its contribution to $|E \cap g(E)|$ is $\sum_i \epsilon_i \epsilon_{i+1}$, indices taken cyclically.

For a transition from current binary state $r \in \{0, 1\}$ to next state $s \in \{0, 1\}$, the matrix entry $M_{rs} = u^s t^{rs}$ is exactly the weight for the next selected vertex and an adjacent selected pair. Consequently the partition function of the cycle is

$$Z_\ell(u, t) = \sum_{\epsilon \in \{0,1\}^\ell} u^{\sum_i \epsilon_i} t^{\sum_i \epsilon_i \epsilon_{i+1}} = \text{tr } M(u, t)^\ell.$$

Different permutation cycles contribute independently to the combinatorial subset generating function. Therefore the following identity is **exact** before truncation:

$$\sum_{E \subseteq [n]} u^{|E|} t^{|E \cap g(E)|} = \prod_{\ell=1}^n Z_\ell(u, t)^{c_\ell(g)}.$$

The determinant and trace of M are $u(t-1)$ and $1+ut$. Thus its characteristic roots are precisely λ_+ from (164) and a second formal root $\lambda_- = u(t-1)/\lambda_+$, satisfying $\lambda_- \in u\mathbb{Q}[t][[u]]$. Cayley–Hamilton, or the standard two-root trace recurrence, gives

$$Z_\ell = \lambda_+^\ell + \lambda_-^\ell.$$

If $\ell > k$, the polynomial/series λ_-^ℓ is divisible by u^ℓ and hence by u^{k+1} . Therefore in the quotient ring modulo u^{k+1} ,

$$Z_\ell \equiv \lambda_+^\ell \quad (\ell > k).$$

Replace every factor corresponding to a long cycle in (168) by λ_+^ℓ modulo u^{k+1} , multiply, and extract the coefficient of u^k . This proves (165).

The dependence on $c_1, \dots, c_k$ and their total contribution to n is now explicit, proving (166). All remaining cycles have lengths at least $k+1$, so their sum is either zero or at least $k+1$. Conversely any such remainder r is realized by one r -cycle, proving (167).

For the LP compression, Theorem 18 permits primal measures invariant under conjugation and dual functions in the orbital-count span. By (166), both their marginal constraints and their dual objective values are constant on any aggregate of conjugacy classes with the same short-cycle counts; all such types are realized by the canonical representatives. Aggregating class masses preserves feasibility and the identity atom, so the sharp LP optimum does not change. The number of nonnegative integer short-cycle vectors is at most $\prod_{\ell=1}^k (1 + \lfloor n/\ell \rfloor) = O_k(n^k)$. QED.

Effective exact arithmetic. This theorem is directly implementable without symbolic eigensolvers. The trace polynomials satisfy

$$Z_0 = 2, \quad Z_1 = 1 + ut, \quad Z_\ell = (1 + ut)Z_{\ell-1} - u(t-1)Z_{\ell-2},$$

while coefficients $\lambda_+ = \sum_{m \geq 0} A_m(t)u^m$, $A_0 = 1$, satisfy the integer-polynomial recurrence

$$A_m = tA_{m-1} - \sum_{i=1}^{m-1} A_i A_{m-i} - \mathbf{1}_{\{m=1\}}(t-1) \quad (m \geq 1).$$

Truncating every polynomial multiplication at u^{k+1} yields (165) using **integer arithmetic only**.

The independent public checker code/check_all_k_orbital_compression.py implements both recurrences using sparse integer dictionaries. It crosschecks the coefficients against an unrelated, direct k -subset enumeration on one representative of **every integer partition** for $3 \leq n \leq 12$ and every $1 \leq k \leq \min(n, 6)$. This finite check is supplementary; the transfer-matrix argument proves (165) for **all** n, k .

Next finite classification frontier. Theorem 30 gives a rigorous compression layer for the exact $k = 4$ primal-dual problem, while Theorem 25 has already resolved its sharp asymptotic coefficient 32. It does not itself give the complete exact finite- n optimum at $k = 4$, which remains a separate classification problem.

26. Complete exact four-subset atom-modulus classification for $4 \leq n \leq 64$

The fixed-rank asymptotic theorem has a substantial finite counterpart in the first previously unresolved rank $k = 4$. Here the exact optimal coefficient can be determined for **every** $4 \leq n \leq 64$, without using numerical optimization as proof.

Theorem 31 (complete degree-64 four-subset spectrum). Let $C_n^{(4)}$ be the sharp marginal-preserving atom/TV coefficient for S_n acting on $\binom{[n]}{4}$. The exact values for all degrees $4 \leq n \leq 64$ are determined by the following disjoint exhaustive cases:

- $n = 4$: $C_4^{(4)} = 1$, because the action is trivial.
- $n = 5$: $C_5^{(4)} = 3/5$, by complementation and Theorem 15 for singleton actions.
- $n = 6$: $C_6^{(4)} = 2/5$, by complementation and the exact two-subset law of Theorem 17.
- $n = 7$: $C_7^{(4)} = 5/14$, by complementation and the exact three-subset certificate in Theorem 19.
- Every $8 \leq n \leq 64$: the exact rational optimum is the value labelled C for that degree in the **fixed public certificate file**

`certificates/four_subset_n8_64.json.`

This includes full exact **attainment**: for each listed degree there is a genuine nonnegative probability measure with all four-subset image marginals uniform and a singleton atom defect exactly $C_n^{(4)}$ times its total variation. Illustrative coefficients are:

n	Exact $C_n^{(4)}$	n	Exact $C_n^{(4)}$
8	5/14	16	3591/9187
9	5/14	20	4147/9871
10	5/14	30	17347/34965
11	1629/4549	40	25257571/45082011
12	131/357	50	297240803/486850773
14	3106/8153	60	261072773/400563233
15	1345/3493	64	6491170033/9749353873

No all-degree exact formula is implied by this bounded, albeit complete, classification.

26.1. Independent exact orbital formulas and full finite search space

For $g \in S_n$, let x, y, z, w be its numbers of cycles of sizes 1, 2, 3, 4, respectively. Put $N = \binom{n}{4}$, and define binomial orbital moments

$$M_j(g) = \sum_{E \in \binom{[n]}{4}} \binom{|E \cap g(E)|}{j}, \quad 0 \leq j \leq 4.$$

One may compute these without transfer matrices:

$$\begin{aligned} M_0 &= N, \\ M_1 &= x \binom{n-1}{3} + (n-x) \binom{n-2}{2}. \end{aligned}$$

Introduce the exact pair-pattern counts

$$\begin{aligned}
c_{22} &= \binom{x}{2} + y, \\
c_{21} &= x(n-x) + (n-x-2y), \\
c_{20} &= \binom{n}{2} - c_{21} - c_{22}.
\end{aligned}$$

For a pair of image-intersection witnesses, the union of the two vertices and their preimages has size 4, 3, 2 according as there are 0, 1, 2 internal directed predecessor edges. Therefore

$$M_2 = c_{20} + (n-3)c_{21} + \binom{n-2}{2}c_{22}.$$

For a triple witness, three internal predecessor edges mean a union of invariant cycles of total length three; two edges mean exactly one path-component or two fixed vertices plus one moved vertex. Directly classifying these cases gives

$$\begin{aligned}
c_{33} &= \binom{x}{3} + xy + z, \\
c_{32} &= \binom{x}{2}(n-x) + x(n-x-2y) + y(n-2-x) + (n-x-2y-3z), \\
M_3 &= c_{32} + (n-3)c_{33}.
\end{aligned}$$

Finally, an invariant four-subset is a union of cycles with total length four, hence

$$M_4 = \binom{x}{4} + \binom{x}{2}y + \binom{y}{2} + xz + w.$$

Binomial inversion gives all five desired orbital counts:

$$F_j^{(4)}(g) = \sum_{i=j}^4 (-1)^{i-j} \binom{i}{j} M_i(g), \quad 0 \leq j \leq 4.$$

Equations (172)–(176) are **integer identities for every permutation** and provide a second derivation independent of the transfer matrix (165). By Theorem 30, the entire class type relevant to these orbitals is represented by

$$(x, y, z, w, r), \quad r = n - x - 2y - 3z - 4w \in \{0\} \cup \{5, 6, \dots\}.$$

Every such type is realized by the permutation with cycles $1^x 2^y 3^z 4^w$ and, if $r > 0$, one additional r -cycle. Conversely every permutation maps to one of these types. Thus verifying (176) and a dual inequality over every tuple (177) is **exhaustive over all** S_n .

26.2. Fixed rational primal-dual certificate proof

For each $8 \leq n \leq 64$, (171) contains two positive rational probability measures P_n, Q_n specified by central conjugacy-class weights and a four-component rational dual vector λ_n , together with two exact rational dual extrema ℓ_n, u_n and the claimed optimum C_n .

The **separate independent verifier**

`code/check_four_subset_n8_64.py`

establishes all of the following as exact integer/rational equalities and inequalities:

1. Both central measures are nonnegative, normalized and **disjointly supported**, with the identity in P_n and not in Q_n .
2. All five orbital moments agree:

$$\mathbb{E}_{P_n} F_j^{(4)} = \mathbb{E}_{Q_n} F_j^{(4)} \quad (0 \leq j \leq 4).$$

Since the orbital relations are exactly the five possible subset-intersection sizes, this implies equality of every individual image marginal.

3. The exact central dual function

$$h_n(g) = \mathbf{1}_{\{g=e\}} - \sum_{j=0}^3 \lambda_{n,j} F_j^{(4)}(g)$$

obeys $\ell_n \leq h_n(g) \leq u_n$ for **every** feasible type (177); no class is omitted. All denominators are cleared before the exhaustive integer comparisons.

4. Every P_n -support class attains u_n , every Q_n -support class attains ℓ_n , and

$$u_n - \ell_n = P_n(e) - Q_n(e) = C_n.$$

5. The size of each canonical conjugacy class is computed exactly and used to exhibit $\delta > 0$ such that $u_{S_n} + \delta(P_n - Q_n)$ remains a nonnegative probability measure. Because the supports are disjoint, this measure has total variation δ , and its identity atom increases by exactly δC_n .

Theorem 18 gives the universal upper bound $C_n^{(4)} \leq u_n - \ell_n$, while this attaining perturbation proves the reverse inequality. Thus the checker gives a **complete finite exact proof** for each of its 57 degrees, conditional only on the mathematical orbital reduction and on replaying the publicly displayed finite integer procedure.

The independent checker calls **no optimizer, floating-point routine, SymPy package or certificate generator**. Its complete published replay checks every feasible compressed conjugacy type for every $8 \leq n \leq 64$, totaling **440,670 types**, and returned:

EXACT k4 FRACTIONAL OPTIMALITY CERTIFIED: 57 DEGREES, 440670 TYPES.

The finite class supports were discovered with numerical linear programming and **reconstructed using exact rational arithmetic before freezing**; that discovery process is not a proof premise. The checker reads only the fixed JSON, recomputes all mathematical constraints independently, and never calls the discovery program. Together with the complement cases $4 \leq n \leq 7$, this establishes Theorem 31. QED.

Scope. This classification is exact but finite. Theorem 25 separately establishes the **all-degree sharp first-order asymptotic** $C_{n^{\{4\}}=1-32/n+O(n^{-2})}$. No exact closed formula for every $n \geq 65$, no globally optimal finite-degree structural classification, and no external human peer review is claimed.